

Lineární algebra 2 - Domácí úkol č. 1

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22. února 2026

Zadání (1.1). Dokažte, že v libovolném komplexním vektorovém prostoru $\mathbf{V}$ se skalárním součinem $\langle \cdot, \cdot \rangle$ platí: (pro libovolné $\mathbf{u}, \mathbf{v} \in \mathbf{V}$)

$$\begin{aligned}\operatorname{Re} \langle \mathbf{u}, \mathbf{v} \rangle &= \frac{1}{4} (\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2), \\ \operatorname{Im} \langle \mathbf{u}, \mathbf{v} \rangle &= \frac{1}{4} (\|\mathbf{u} - i\mathbf{v}\|^2 - \|\mathbf{u} + i\mathbf{v}\|^2),\end{aligned}$$

kde $\operatorname{Re}(x)$ značí reálnou a $\operatorname{Im}(x)$ imaginární část komplexního čísla x . (Pro inspiraci vizte tvrzení 8.32 ve skriptech.)

K výpočtu použijeme následující vlastnosti:

- $z + \bar{z} = 2 \operatorname{Re} z, z \in \mathbb{C}$
- $z - \bar{z} = 2i \operatorname{Im} z, z \in \mathbb{C}$
- $\|\mathbf{u}\|^2 = \langle \mathbf{u}, \mathbf{u} \rangle$
- Pak vlastnosti SSS, SL1, SL2 a vlastnosti (1), (2), (3) z pozorování 8.15.

$$\begin{aligned}\operatorname{Re} \langle \mathbf{u}, \mathbf{v} \rangle &= \frac{1}{4} (\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2) = \frac{1}{4} (\langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle - \langle \mathbf{u} - \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle) \\ &= \frac{1}{4} (\langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} \rangle - (\langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle - \langle \mathbf{u}, \mathbf{v} \rangle - \langle \mathbf{v}, \mathbf{u} \rangle)) \\ &= \frac{1}{4} (2\langle \mathbf{u}, \mathbf{v} \rangle + 2\langle \mathbf{v}, \mathbf{u} \rangle) = \frac{1}{2} (\langle \mathbf{u}, \mathbf{v} \rangle + \overline{\langle \mathbf{u}, \mathbf{v} \rangle}) = \frac{1}{2} (2 \operatorname{Re} \langle \mathbf{u}, \mathbf{v} \rangle) = \operatorname{Re} \langle \mathbf{u}, \mathbf{v} \rangle \quad \square\end{aligned}$$

$$\begin{aligned}\operatorname{Im} \langle \mathbf{u}, \mathbf{v} \rangle &= \frac{1}{4} (\|\mathbf{u} - i\mathbf{v}\|^2 - \|\mathbf{u} + i\mathbf{v}\|^2) = \frac{1}{4} (\langle \mathbf{u} - i\mathbf{v}, \mathbf{u} - i\mathbf{v} \rangle - \langle \mathbf{u} + i\mathbf{v}, \mathbf{u} + i\mathbf{v} \rangle) \\ &= \frac{1}{4} (\langle \mathbf{u}, \mathbf{u} \rangle - \langle \mathbf{v}, \mathbf{v} \rangle - \langle \mathbf{u}, i\mathbf{v} \rangle - \langle i\mathbf{v}, \mathbf{u} \rangle - (\langle \mathbf{u}, \mathbf{u} \rangle - \langle \mathbf{v}, \mathbf{v} \rangle + \langle \mathbf{u}, i\mathbf{v} \rangle + \langle i\mathbf{v}, \mathbf{u} \rangle)) \\ &= -\frac{1}{2} (\langle \mathbf{u}, i\mathbf{v} \rangle + \langle i\mathbf{v}, \mathbf{u} \rangle) = -\frac{1}{2} (i\langle \mathbf{u}, \mathbf{v} \rangle - i\langle \mathbf{v}, \mathbf{u} \rangle) = -\frac{i}{2} (\langle \mathbf{u}, \mathbf{v} \rangle - \overline{\langle \mathbf{u}, \mathbf{v} \rangle}) = -\frac{i}{2} (2i \operatorname{Im} \langle \mathbf{u}, \mathbf{v} \rangle) = \operatorname{Im} \langle \mathbf{u}, \mathbf{v} \rangle \quad \square\end{aligned}$$